Related Work & Research Gaps

System	Venue	Multi-agent Router	VLM Disease	Soil Sensor	Weather Prediction	XAI Farmer UI	Voice / Indic	Market Price	Deployed
Farmer.Chat [?]	ACM 2024	×	× (Generic)	×	✓ API	×	✓ Partial	×	✓ 565k users
AgroAskAI [?]	arXiv 2024	✓ Chain	×	×	✓ Dedicated	×	✓ Partial	×	×
AgriGPT [?]	arXiv 2025	✓ Data engine	✓ VL ext.	×	✓ Partial	×	×	×	×
AgriM-LLM [?]	Preprints 2026	×	✓ Best VE	×	×	×	×	×	×
AgriDoctor (Base) [?]	IEEE 2025	✓ Router+5	✓ Strong	×	×	×	✓ CN/EN only	×	×
AgriBuddy [?]	BRAC 2025	✓ Partial	✓ CNN rice	×	✓ Partial	×	×	×	✓ Partial
PhenoAssistant [?]	Nature 2026	✓ Yes	✓ Research	×	×	×	×	×	× (Lab)
Agentic AI [?]	MDPI 2026	✓ 4 agents	✓ Rice CNN	✓ Partial	✓ Dedicated	×	×	×	×

Key Research Gaps Identified

- **Soil Sensor Fusion** and **Market Price** features are completely missing from existing systems.
- No prior platform supports explainable AI (**XAI**) for farmer UIs or comprehensive **Indic speech** interfaces.

System Architecture

